Shreemit Garimella

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Skills Summary

- **Languages** Python, Java, JavaScript, Swift, SQL, C++, C
- **Frameworks** PyTorch, Scikit, NLTK, TensorFlow, Keras, Flask, Airflow, Pandas, Matplotlib, HuggingFace, OpenAI
- **Tools** Git, Docker, Jira, AWS, Azure, Weights and Biases, MongoDB, MySQL, Tableau, SQLite

Education

- **University of Washington** Seattle, USA
 Master of Science in Electrical and Computer Engineering *Sept 2022 - Aug 2024*
 Notable Courses: *Biomedical Applications of LLMs, Natural Language Processing, Deep Learning, TinyML, Explainable AI*
- **Visvesvaraya Technological University (BMSIT)** Bangalore, India
 Bachelor of Engineering in Electronics and Communication Engineering *Aug 2017 - July 2021*
 Notable Courses: *Python Application Programming, Artificial Neural Networks, Operating Systems.*

Experience

- **University of Washington** Seattle, WA
 Teaching Assistant: LLMs course *Dec 2023 - March 2024*
 - Designed and facilitated engaging coding assignments and mini-projects, applying LLMs to real-world scenarios and leveraging interactive discussions and collaborative activities to solidify student learning.
- **Global Health Labs** Bellevue, WA
 Machine Learning Research Intern *June 2023 - Sept 2023*
 - Trained computer vision ML models by conducting an ablation study and building MLOps pipelines using Weights and Biases to enable fast experimentation and comparisons for cervical cancer screening achieving an accuracy of 94%
 - Developed an embedding projector tool utilizing semantic maps using GradCAM to visualize data interactions within class labels and improve the cervical cancer classification model's predictions
- **NanoString** Seattle, WA
 Machine Learning Capstone *Jan 2023 - June 2023*
 - Trained a custom deep learning network based on U-Net architecture using PyTorch, for RNA within tissue samples accurately, achieving a prediction accuracy improvement of 40% compared to traditional image processing methods
 - Expedited the identification of RNA molecules across four color emission PSFs, by employing a transfer learning strategy, resulting in reduced training time and computational resources
- **HumanFractal.ai** Bangalore, India
 Software Developer *March 2021 - June 2022*
 - Designed and created a mobile application, Resolute which facilitated secure EHR data storage and offered AI-powered preventive care, leading to over 8000 active users and a 4.7-star rating
 - Developed a machine learning model to anticipate and recommend preventive measures based on user behavior through comprehensive data analysis and preprocessing, using Pandas and SQL, along with model training, validation, and deployment, optimizing the model's performance
 - Created secure and privacy-preserving services for storing patient records in DynamoDB through REST APIs using AWS Lambda and Amplify which allowed the company to develop trust with their users and their data
 - Deployed user authentication, login, and sign-up using Amazon Cognito to improve the ability to maintain security and privacy, and increased the time and budget savings for the startup
- **HumanFractal.ai** Bangalore, India
 Software Engineer Intern *Nov 2020 - Feb 2021*
 - Developed an interactive chatbot using Amazon Lex and AWS Lambda to streamline hospital appointment scheduling and enhance customer experience with a cutting-edge virtual assistant solution that reduced hospital workloads by 20%.
 - Optimized scalability and reliability by integrating serverless Lambda functions to fetch appointments through RESTful API, leveraging AWS serverless architecture for seamless scaling and cost-effectiveness.

Projects

- **Optimizing Vision Transformer Mixture of Experts for CIFAR-10 Classification**: Developing an advanced Vision Transformer (ViT) Mixture of Experts model for computer vision classification, optimizing performance on the CIFAR-10 dataset, with a focus on enhancing accuracy and efficiency through dynamic expert selection.
- **LeaseGPT**: Developed a personalized chatbot utilizing OpenAI LLMs, Langchain, and Pinecone Vector DB, serving as a virtual leasing agent for users searching for apartments tailored to their specific needs based on Retrieval Augmented Generation strategy
- **Recipe Generator**: Fine-tuned seq2seq T5 on a dataset of over 1 million recipes and generated novel recipes given an ingredients list using Pytorch, Hugging Face Transformers, and Pandas fine-tuning my skills around Natural Language Processing Techniques
- **Handwritten notes captioning**: Finetune a multimodal machine learning model for handwritten note recognition using Low-Rank Adaptation (LoRA), enabling efficient annotation of student assignments.
- **Movie Recommendation System**: Developed a movie recommendation system using Python, leveraging both collaborative and content-based filtering methods; evaluated system performance using precision, recall, and accuracy metrics.

Certifications and Awards

- AI-900 Azure AI Fundamentals, Neural Networks and Deep Learning, Structuring Machine Learning Projects
- Awarded 2nd place in the Engine Capstone Competition at the University of Washington